2. **SOLUTIONS 2017**

$$E(aX_1 + bX_2) = aE(X_1) + bE(X_2)$$
$$= a\mu + b\mu$$
$$= \mu \quad \text{if unbiased.}$$

$$\Rightarrow \quad (a+b)\mu = \mu \quad \Rightarrow \quad a+b=1$$

$$MSE(aX_1 + bX_2) = var(aX_1 + bX_2)$$
$$= a^2 var(X_1) + b^2 var(X_2)$$
$$= a^2 + 2b^2$$
$$= a^2 + 2(1-a)^2$$
$$= a^2 + 2 - 4a + 2a^2$$
$$= 3a^2 - 4a + 2$$

Diff. w.r.t. a $\quad \frac{dMSE}{da} = 6a - 4$

Equate to 0 $\Rightarrow \quad 6a = 4 \Rightarrow \quad a = \frac{2}{3}$

3. $Bias(a\bar{X}) = E(a\bar{X}) - p$
$= aE(\bar{X}) - p$
$= aE(X) - p$
$= ap - p$
$= (a-1)p$

$MSE(a\bar{X}) = Bias^2(a\bar{X}) + var(a\bar{X})$
$= (a-1)^2 p^2$

$var(a\bar{X}) = a^2 var(\bar{X}) = a^2 \frac{var(X)}{n}$
$= a^2 \frac{p(1-p)}{n}$

(12) 4. In 8 weeks : 7 patients attack.

Consider 5 patients who have not had an attack.

$$\int_8^\infty \lambda e^{-\lambda t}\, dt$$

$$\left[-e^{-\lambda t} \right]_8^\infty$$

$$= 0 + e^{-8\lambda} = \boxed{e^{-8\lambda}}$$

So for all 5 we have $(e^{-8\lambda})^5 = e^{-40\lambda}$

$$L(\lambda) = \prod_{i=1}^{7} \lambda e^{-\lambda t_i} \cdot e^{-40\lambda}$$

$$= \lambda^7 e^{-\sum_{i=1}^{7} t_i \lambda} e^{-40\lambda}$$

$$= \lambda^7 e^{-\lambda\left(\sum_{i=1}^{7} t_i + 40\right)}$$

$$l(\lambda) = 7\log\lambda - \lambda\left(\sum t_i + 40\right)$$

5. $\bar{x} \pm t_{0.025,4} \frac{s}{\sqrt{n}}$

$\bar{x} \pm 2.776 \times \frac{s}{\sqrt{5}}$ → D

6. See Tutorial Sheet 2

1. C.